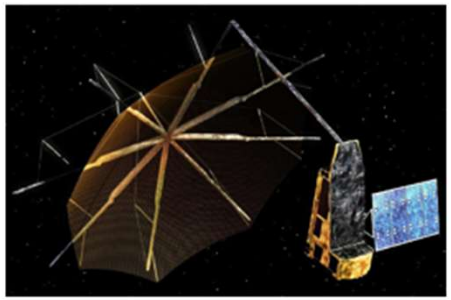
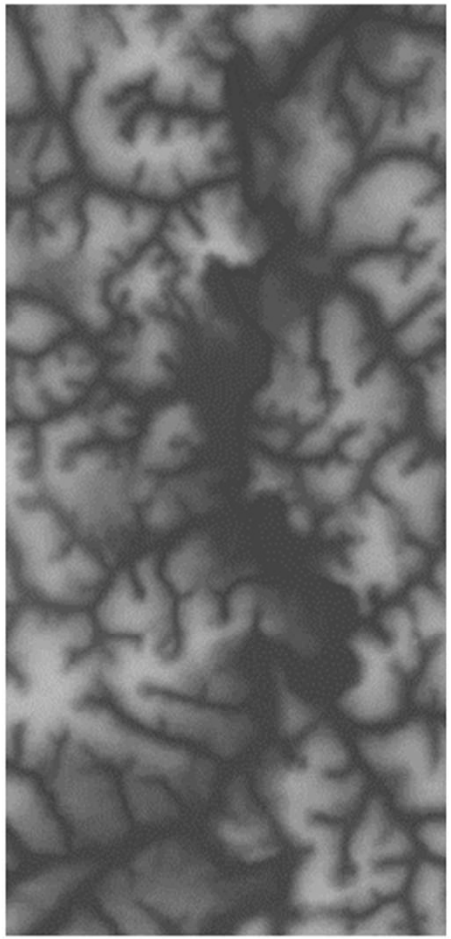
EEBIOMASS Virtual Workshop on BIOMASS CALVAL

Session 2 – 29 October 2026
Beyond the Trees: Transponder Calibration, Ionospheric Corrections, and Secondary Mission Objectives





Detecting topography and water under tropical forest with P-band radar

Laurent Polidori
UFPA (Belém, Brazil)

ESA Project PP0104563
"AMAZON3D CALVAL: Validation of Biomass-derived DTM under Amazon rainforest"
Laurent Polidori, Mhamad El Hage, Laurent Ferro-Famil, Ludovic Villard, Carlos Caldeira

Challenge 1: the Amazon is resistant to cartographic techniques

Huge area and access difficulties

Poor geodetic infrastructure

Atmospheric effects: troposphere (clouds), ionosphere

Sun-synchronous orbits: longer revisit period in the tropics

Altimetry (IGBE)



RBMC (IBGE)





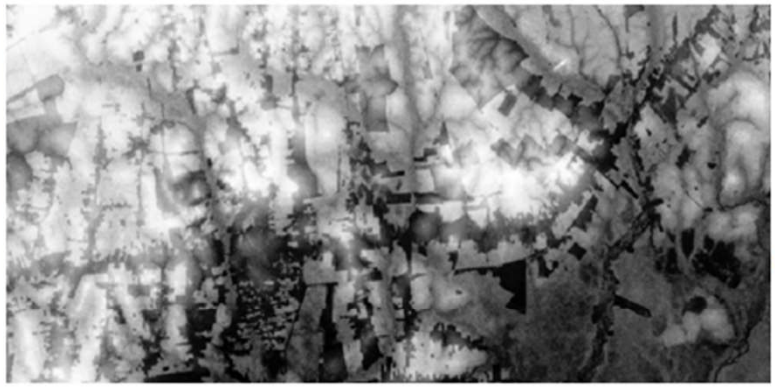
Challenge 2: most Digital Elevation Models are canopy models (DSMs)

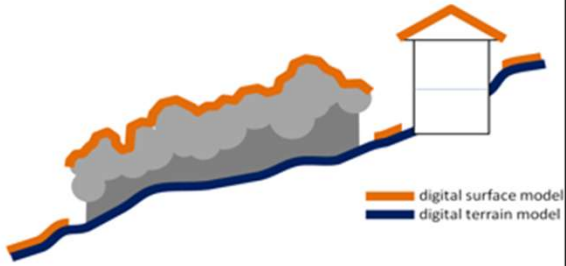
But most users need a terrain model (DTM)

Landsat



SRTM





Elevation error

Example: underestimation of water volume in a dam simulation

Slope error

Scale-dependente error (the higher the resolution, the lower the accuracy)

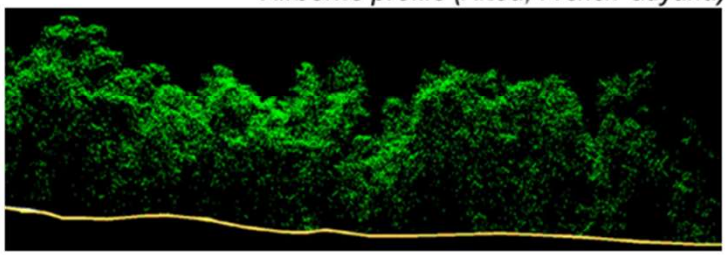
data base	grid mesh size (m)	expected slope error (°)
SRTM	90	2,0
ASTER GDEM	30	5,0

Polidori L., Simonetto E. (2014) Effect of scale on the correlation between topography and canopy elevations in an airborne InSAR product over Amazonia. *Procedia Technology* 16, 180-185,

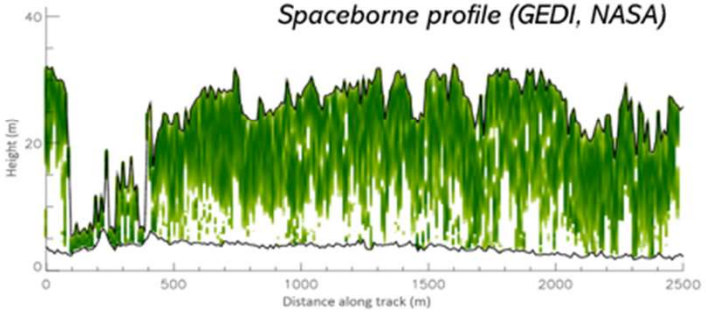
Strategies for topographic modeling through forest

1. Lidar

Airborne profile (Altoa, French Guyana)

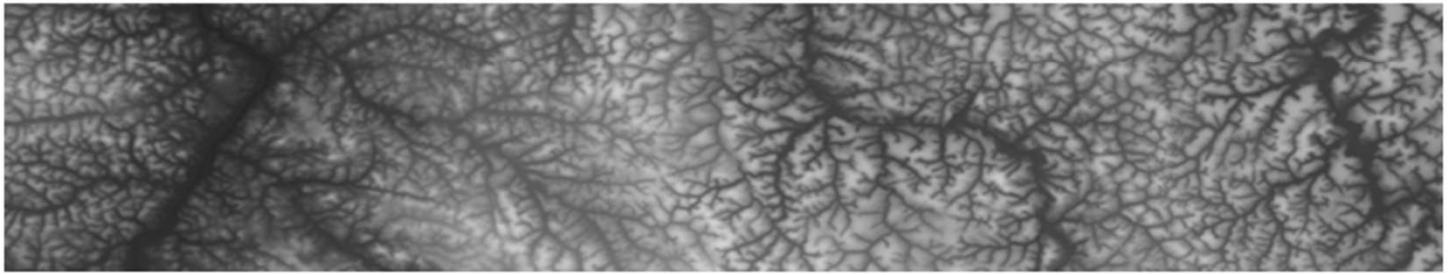


Spaceborne profile (GEDI, NASA)



2. P-band radar

Airborne P-band InSAR (Amapá, Brazil)



3. "Intelligent" post-processing of DSM (e.g. FABDEM, ANADEM...)

- **Compile studies on the potential of P-band radar for topography through forests**
- **Prepare validation of Biomass product with relevant quality criteria (wrt user needs)**

Polidori, L.; Caldeira, C.R.T.; Smessaert, M.; El Hage, M. 2022. Digital elevation modeling through forests: the challenge of the Amazon. *Acta Amazonica* 52: 69-80. DOI: 10.1590/1809-4392202103091.

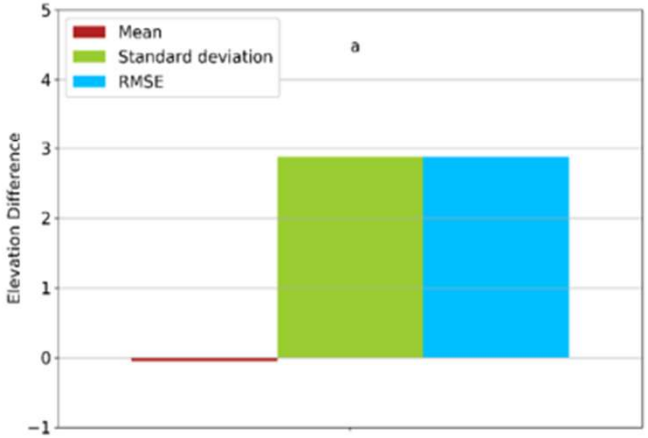
Biomass CALVAL campaign (Paracou, French Guiana)

Confirmation of the potential of P-band SAR to separate the ground and the forest

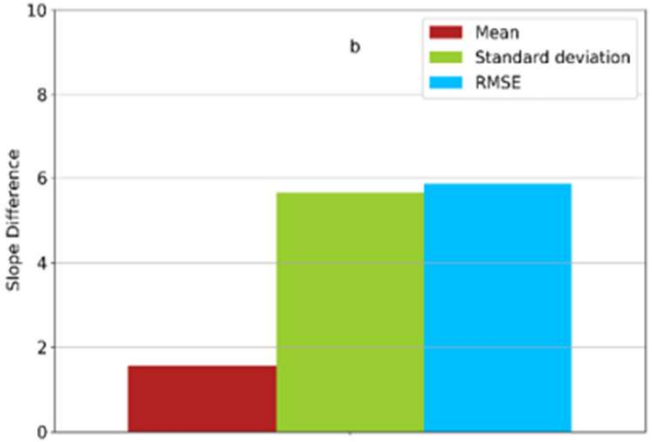
External validation with LiDAR survey as a reference:

- No significant bias
- 125 MHz : height RMSE = 2 m
- 6 MHz : height RMSE = 3 m

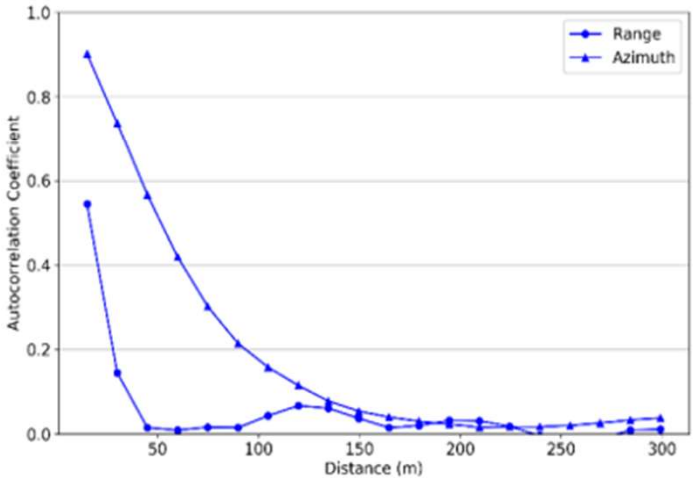
elevation error



slope error



error autocorrelation



El Hage M., Villard L., Huang Y., Ferro-Famil L., Koleck T., Le Toan T., Polidori L. (2022) Multicriteria accuracy assessment of DEMs produced by airborne P-band polarimetric SAR tomography in tropical rainforests. *Remote Sensing*, 14:4173.

El Hage M., Villard L., Ferro-Famil L., Polidori L. (2024) Potential of the upcoming Biomass P-band radar mission for digital terrain modelling beneath dense tropical forests: first accuracy assessment. *International Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, v. X-3-2024, 125-130.

Limitations of the experiment

Reference data
Considering LiDAR as a perfect reference is highly questionable.
Acceptable accuracy for elevations given the desired accuracy.
Insufficient accuracy for slopes due to the contribution of the interpolation process and the variable density of ground points.

Solution:
reduce dependence on reference data by implementing internal validation, which consists in searching for inconsistencies in the DTM.

Topographic metrics
Evaluating elevations alone is not representative of the quality of a DTM.
The fidelity of the model to terrain shapes depends on the accuracy of the elevation derivatives (slope, aspect...)

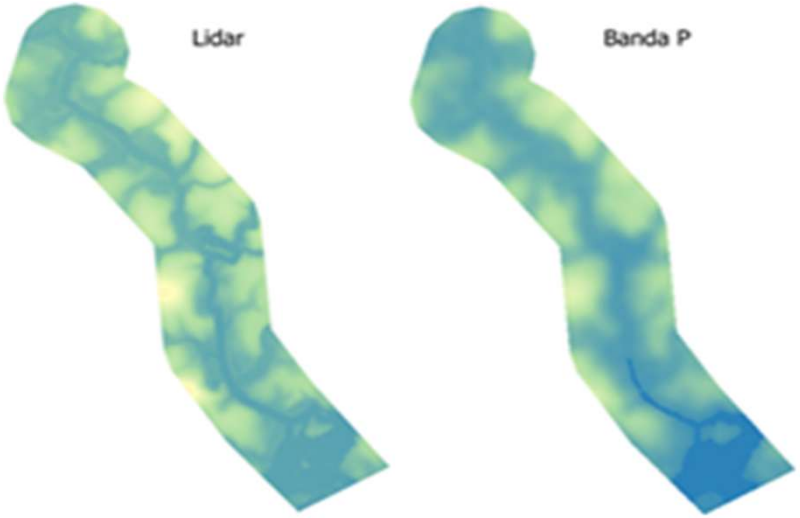
Solution:
evaluate slope accuracy
evaluate error autocorrelation
(compliance with terrain shapes)

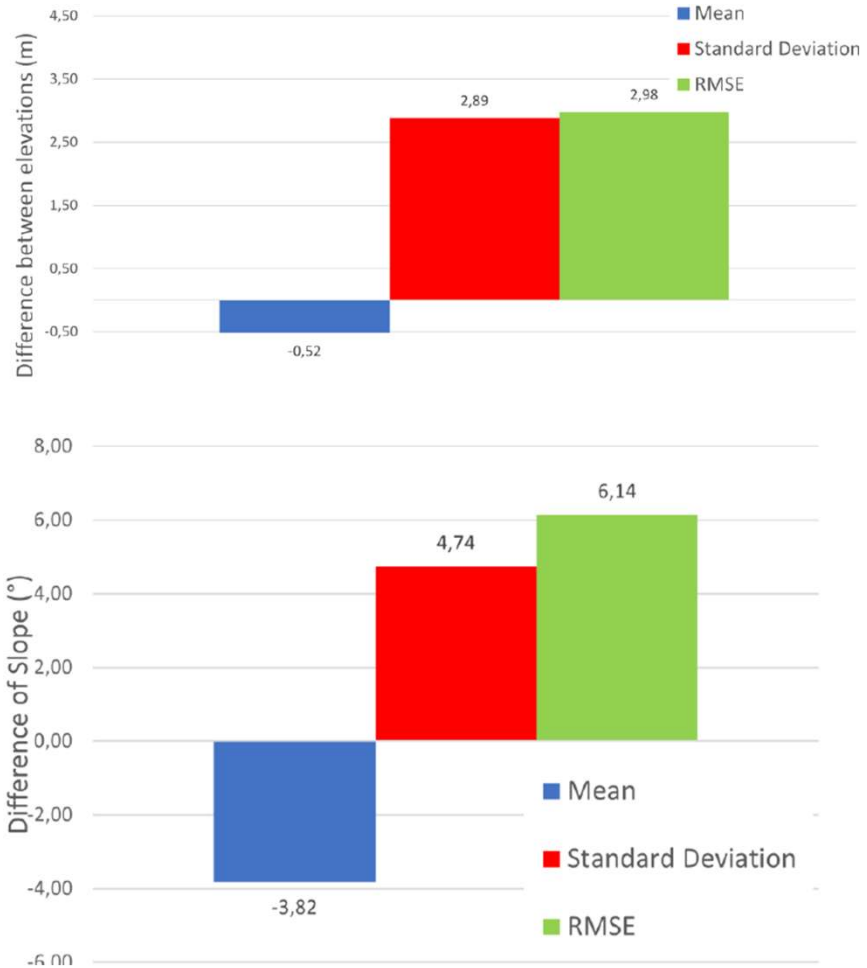
Spatial extension
The size of the observed area does not allow access to the long wavelengths of the relief.
The area does not contain a complete watershed of significant size to evaluate the hydrographic consistency of the DEM

Solution:
test on a larger area
(Amapá, Brazilian Amazon).

Validation of P-band InSAR survey (Amapá, Brazil)

- **external validation**



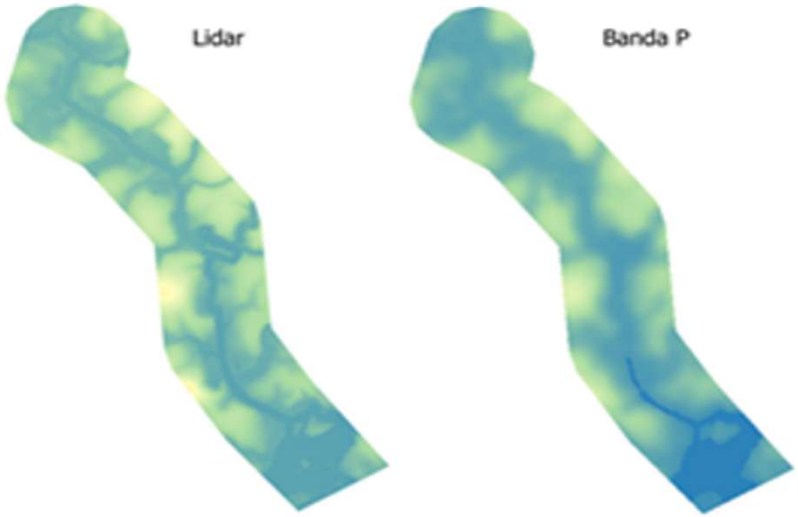


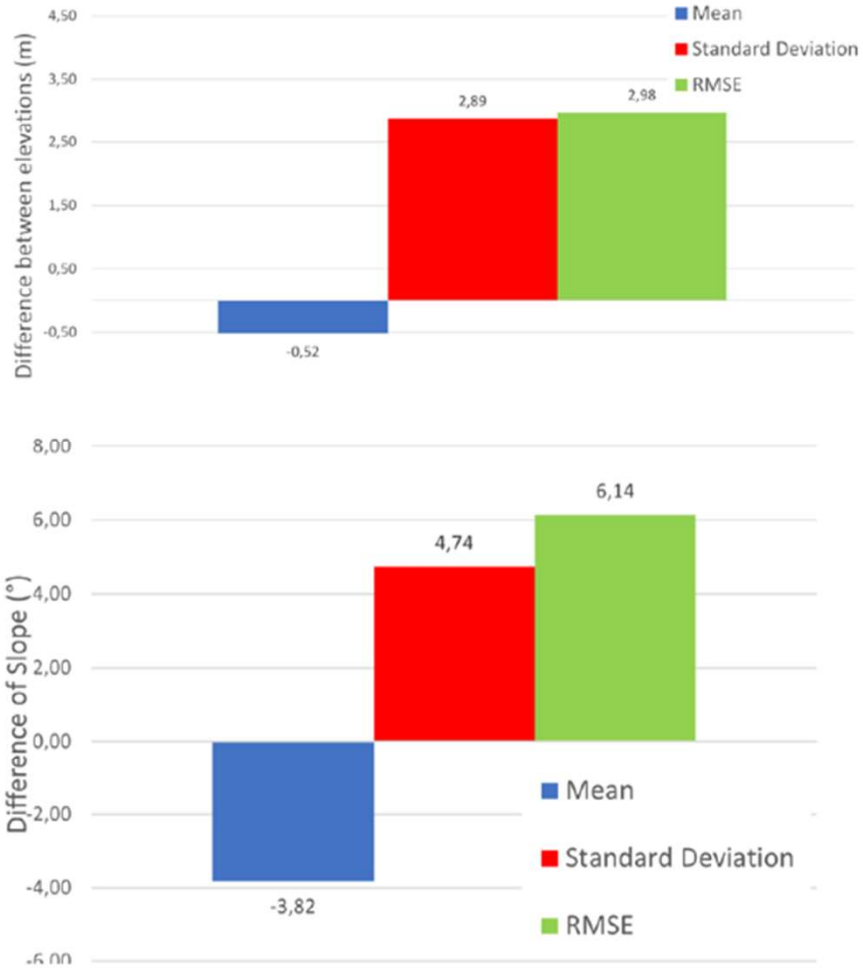
Caldeira, C.R.T.; El Hage, M.; Caldeira, M.C.O.; Gorgens, E.B.; Ometto, J.P.H.B; Polidori, L. 2023. Comparison between Digital Terrain Models generated by P-Band Radar and LiDAR in the Amazon, a case study in Amapá (Brazil). *Northeast Geosciences Journal*, vol. 9, 1:59-70. DOI: 10.21680/2447-3359.2023v9n1ID31574.

Caldeira, C.R.T.; El Hage, M.; Rosa, R.A.S.; Polidori L. 2024. Accuracy and hydrographic realism of an under- forest DEM derived from airborne P-band radar interferometry over a wide area in the Brazilian amazon, *International Journal of Remote Sensing*, 45:16, 5295-5316, DOI: 10.1080/01431161.2024.2372075.

Validation of P-band InSAR survey (Amapá, Brazil)

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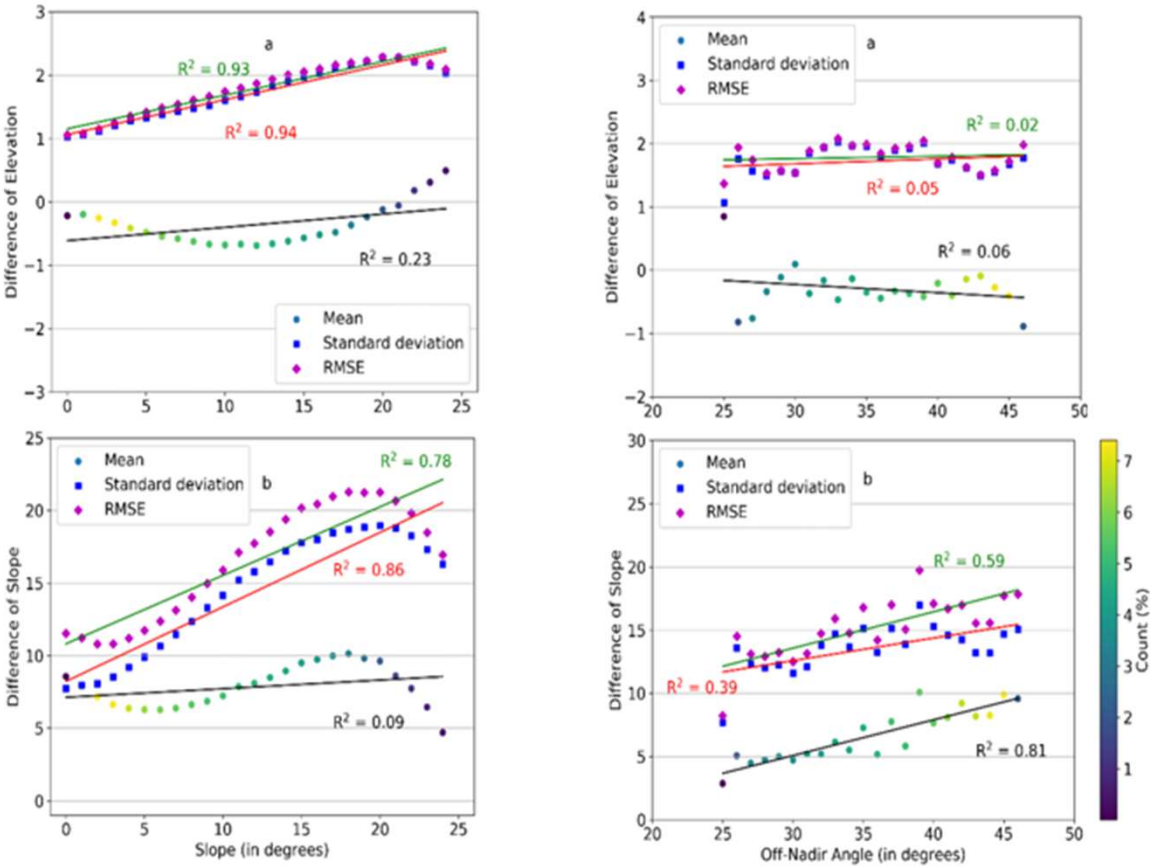




Caldeira, C.R.T.; El Hage, M.; Caldeira, M.C.O.; Gorgens, E.B.; Ometto, J.P.H.B; Polidori, L. 2023. Comparison between Digital Terrain Models generated by P-Band Radar and LiDAR in the Amazon, a case study in Amapá (Brazil). *Northeast Geosciences Journal*, vol. 9, 1:59-70. DOI: 10.21680/2447-3359.2023v9n1ID31574.

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Sensitivity analysis



Application to operational cartography

Validation of Biomass products
Densification strategies (90 to 30 m)
User need consolidation
Technical specification with mapping agencies

Hydrology through forest

Detection and mapping of flooded forest (várzea and igapó) in Central Amazon, by integrating polarimetric descriptors (entropy, alpha and double-bounce), radar altimetry and land use and land cover (internship UFPA/CESBIO)

Quaternary geology

Analysis of terrain features with low altimetric amplitude under forest cover